Li Zhang

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Personal Website · GitHub

Education

Hong Kong University of Science and Technology Sep 2023 – Jun 2027 (Expected)
B.Eng. in Computer Science GPA: 3.86/4.30 Class Rank: 22/170

- **Selected coursework:** Large-Scale Machine Learning System for Foundation Models; Advanced Computer Architecture; Modern Compiler Construction; Computer Communication Networks; Design and Analysis of Algorithms
- **Honors:**
  - HKUST Dean's List, School of Engineering, Fall 2023, Spring 2024, Spring 2025, Fall 2025 and Spring 2026
  - University Scholarship Scheme for Continuing Undergraduate Students, 2024–25 and 2025–26 (HK$ 10,000/year)

ETH Zurich Sep 2026 – Jan 2027 (Expected)
Exchange Student, Department of Computer Science

- **Current coursework:** Large-Scale AI Engineering; Design of Parallel and High-Performance Computing; Computer Systems; Hands-On Deep Learning

Research Interests

Machine learning systems for foundation model training and inference, particularly distributed runtimes, scheduling and resource management, performance modeling, and hardware-aware optimization.

Research Experience

LLM Serving Simulation for Hierarchical KV-Cache Systems Feb 2026 – Present
Mentored by Shiyi Liu, Ph.D. student advised by Prof. Yuan Xie

- Developed a vLLM platform plugin for request-trace recording and replay, enabling reproducible serving experiments without executing real model computations.
- Extended vLLM-Sim to support LMCache-based hierarchical KV-cache systems by replacing physical GPU/CPU memory operations with simulated storage and modeling CPU-to-GPU transfer latency using profiled bandwidth.
- Improved the modeling of asynchronous serving behavior by aligning backend execution with vLLM scheduling semantics and refining kernel profiling for pipelined and mixed prefill/decode execution.
- Evaluating simulator fidelity for LMCache-enabled serving across workloads and memory configurations, using TTFT, TPOT, and end-to-end latency against real-system measurements.

LLVM SSA Construction and Indirect-Call Benchmark Generation May 2025 – Dec 2025
Supervised by Prof. Jiasi Shen

- Implemented an LLVM SSA transformation pass based on Cytron et al., including dominator analysis, dominance-frontier-based ϕ-function placement, and variable renaming.
- Built a compile-validated program-transformation pipeline generating C++ benchmarks with diverse indirect-call patterns for WebAssembly analysis, using LLM-guided transformations and automated repair.

Selected Course Projects

Distributed AI Training and Communication on Alps Supercomputer Sep 2026 – Present
ETH Zurich · Large-Scale AI Engineering (Graduate Level)

- Implemented data, tensor, and pipeline parallelism in PyTorch, including custom autograd operations, inter-stage communication, and correctness validation against non-parallel baselines.
- Benchmarked NCCL AllReduce across four nodes and 16 GPUs on the Alps supercomputer, evaluating achieved communication bandwidth across different message sizes and process-group topologies.
- Evaluated Transformer training performance under different compilation, optimizer, and sequence-length configurations, and profiled GPU kernel execution using NVIDIA Nsight Systems.

Hardware–Software Co-Optimization of DNN Kernels with Gemmini Mar 2026 – May 2026
ELEC 5140 Advanced Computer Architecture (Graduate Level)

- Explored Gemmini accelerator configurations and software tiling strategies across CNN and MLP workloads to characterize performance scaling and hardware-resource bottlenecks.
- Analyzed execution, memory-transfer, TLB-wait, and reservation-station counters to characterize workload-specific bottlenecks and explain performance saturation.
- Achieved a 4.0× simulated speedup over baseline on a convolution workload through joint hardware scaling and software tiling, while identifying diminishing returns from additional accelerator resources.

Amy Compiler Feb 2026 – May 2026

COMP 4121 Modern Compiler Construction

- Implemented an interpreter and a compiler for the Amy functional language in Scala, including regex-based lexing, LL(1) parsing, constraint-based type checking via unification, and WebAssembly code generation.

Publications

1. D. Liu, Y. Wang, H. C. Iu, **L. Zhang**, C. Shi, Y. Lu, X. Ma. "CoNarrate: Computationally Modeling Information Prematureness and Scaffolding the Co-Construction of Patient Narratives in Asynchronous Telehealth." *ACM Symposium on User Interface Software and Technology (UIST 2026)*, to appear.
2. K. Yuan, **L. Zhang**, H. Lyu, Z. Pan, Y. Zhang, J. Li, B. Guo, J. Hu, Q. Guo, X. Ma. "I Love the Internet Again: Exploring the Interaction Inception of 'TikTok Refugees' Flocking into RedNote." *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA 2025)*. doi:10.1145/3706599.3719738

Professional Experience

Undergraduate Teaching Assistant Dec 2025 – Feb 2026

COMP 2611 Computer Organization, supervised by Prof. Cindy Xin Li

- Developed prototype digital-logic labs using 74-series ICs, covering combinational building blocks including adder, multiplexer and ALU with LED-based signal visualization.
- Prepared curriculum-development reports benchmarking computer organization and computer architecture courses and competition-based extensions for future course design.

Trainee IT Engineer Dec 2024 – Jan 2025

Lenovo (Beijing) Limited

- Developed and deployed an internal reporting dashboard for the Order Service Platform using Microsoft Power BI, integrating data from a PostgreSQL database to support business reporting and visualization.

Skills

Programming: C/C++, Python, Scala
ML & Distributed Systems: PyTorch, vLLM, LMCache, NCCL
Compilers & Architecture: LLVM, Gemmini (Chipyard)
Systems & Tools: Linux, xv6 (RISC-V), Slurm, GDB, Git
Languages: Mandarin Chinese (Native), English (IELTS 7.5), Cantonese (Basic)

Leadership & Community Engagement

The Dispersions, Student Band Jan 2023 – Present

- Co-founded the band; serve as rhythm guitarist and co-lead vocalist, write original songs, and perform at campus events, band competitions, live music venues, and music festivals in Hong Kong and Zhuhai.

HKUST Peer Companion Dec 2024 – Present

- Supported campus wellness activities, including peer coffee chats, stress-relief events, and exam-week programs, reaching hundreds of students.

HKUST Exchange Buddy Jan 2026 – Jun 2026

- Supported incoming exchange students in adapting to campus life and building cross-cultural connections through dinners, coffee chats, and campus and city tours.

香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Name: ZHANG, Li 张力
Student ID: 21041813
HK ID: F361256(9)
Date of Birth: 14 Nov 2004
Gender: Male

Transcript of Academic Record

Date of Issue: 17 September 2026

Issue To: ZHANG, Li

Academic Program

Admit Date: 1 September 2023
Program: Bachelor Degree in School of Engineering (with Extended Major in Artificial Intelligence)

Program Change: 2024-25 Fall
Program: Bachelor of Engineering
Major: Computer Science (with Extended Major in Artificial Intelligence)

Program Change: 2024-25 Fall
Program: Bachelor of Engineering
Major: Computer Science

Study Abroad: 2026-27 Fall
Swiss Federal Institute of Technology Zurich

Academic Records

2023-24 Fall (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
COMP1021	Introduction to Computer Science	3.0	3.0	A
COMP2711H	Honors Discrete Mathematical Tools for Computer Science	4.0	4.0	B+
ELEC1100	Introduction to Electro-Robot Design	4.0	4.0	A
HMAW1905B	Behavioral Foundations of University Education: Habits, Mindsets, and Wellness	3.0	-	PP
LANG1404	Advanced Academic English for University Studies	3.0	3.0	A
LEGL1000	Legal Education (PRC Constitution, Basic Law, National Security Law and Legal System of HKSAR)	0.0	-	P
MATH1013	Calculus IB	3.0	3.0	A+

TGA: 3.888

Transfer Credit(s) from Pre-tertiary Qualifications and equivalent course(s) of HKUST:

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
LANG1402	English Language for University Studies	3.0	3.0	T

CGA: 3.888

Cumulative Credits Earned: 17.0
Cumulative Transfer Credits: 3.0

Awards: Dean's List

2023-24 Winter (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
COMP1029C	C Programming Bridging Course	1.0	1.0	P

TGA: **
CGA: 3.888

Cumulative Credits Earned: 18.0
Cumulative Transfer Credits: 3.0

2023-24 Spring (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
CHEM1020	General Chemistry I	3.0	3.0	A
COMP2011	Programming with C++	4.0	4.0	A
COMP2211	Exploring Artificial Intelligence	3.0	3.0	B
HMAW1905B	Behavioral Foundations of University Education: Habits, Mindsets, and Wellness	3.0	3.0	P
IEDA2010	Introduction of Industrial Engineering and Decision Analytics	3.0	3.0	A-
MATH1014	Calculus II	3.0	3.0	A-
PHYS1112	General Physics I with Calculus	3.0	3.0	A

TGA: 3.747
CGA: 3.814

Cumulative Credits Earned: 40.0
Cumulative Transfer Credits: 3.0

Academic Registrar

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香港科技大學
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Name: ZHANG, Li 张力
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Gender: Male

Transcript of Academic Record

Awards: Dean's List

2023-24 Summer (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
UROP1000	Undergraduate Research Opportunities	0.0	-	P

TGA: **
CGA: 3.814

Cumulative Credits Earned: 40.0
Cumulative Transfer Credits: 3.0

2024-25 Fall (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
COMP2012	Object-Oriented Programming and Data Structures	4.0	4.0	A-
COMP2611	Computer Organization	4.0	4.0	B+
COMP4900	Academic and Professional Development	0.0	-	PP
ENGG2010	Engineering Seminar Series	0.0	-	PP
MATH2011	Introduction to Multivariable Calculus	3.0	3.0	A
MATH2111	Matrix Algebra and Applications	3.0	3.0	B+
UROP1100N	Undergraduate Research Opportunities Series 1	1.0	1.0	P

TGA: 3.564

Transfer Credit(s) from Nanyang Technological University through virtual study and equivalent course(s) of HKUST:

HUMA1920	Introduction to Moral Philosophy	3.0	3.0	T

CGA: 3.744

Cumulative Credits Earned: 55.0
Cumulative Transfer Credits: 6.0

Awards: University's Scholarship Scheme for Continuing Undergraduate Students

2024-25 Winter (Full-Time)

Awards: University's Scholarship Scheme for Continuing Undergraduate Students

2024-25 Spring (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
COMP1991	Industrial Experience	0.0	-	PP
COMP3711	Design and Analysis of Algorithms	3.0	3.0	A-
COMP4611	Design and Analysis of Computer Architectures	3.0	3.0	A-
COMP4900	Academic and Professional Development	0.0	-	PP
ECON1221	Thinking like an Economist II: Macroeconomics	3.0	3.0	A+
ECON2103	Principles of Microeconomics	3.0	3.0	A+
ELEC2600	Probability and Random Processes in Engineering	4.0	4.0	A
ENGG2010	Engineering Seminar Series	0.0	-	P
UROP1100O	Undergraduate Research Opportunities Series 1	1.0	1.0	P

TGA: 4.000
CGA: 3.806

Cumulative Credits Earned: 72.0
Cumulative Transfer Credits: 6.0

Awards: Dean's List
University's Scholarship Scheme for Continuing Undergraduate Students

2024-25 Summer (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
COMP1991	Industrial Experience	0.0	-	PP
UROP1100P	Undergraduate Research Opportunities Series 1	1.0	1.0	P

TGA: **
CGA: 3.806

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THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Name:	**ZHANG, Li 张力**		
Student ID:	**21041813**	**Date of Birth:**	**14 Nov 2004**
HK ID:	**F361256(9)**	**Gender:**	**Male**

Transcript of Academic Record

Cumulative Credits Earned: 73.0
Cumulative Transfer Credits: 6.0

Awards: University's Scholarship Scheme for Continuing Undergraduate Students

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2025-26 Fall (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
COMP1991	Industrial Experience	0.0	-	PP
COMP2633	Competitive Programming in Cybersecurity I	0.0	-	P
COMP3031	Principles of Programming Languages	3.0	3.0	B+
COMP3631	Cryptography	3.0	3.0	A+
COMP4900	Academic and Professional Development	0.0	-	PP
ELEC3120	Computer Communication Networks	3.0	3.0	A+
PHYS1114	General Physics II	3.0	3.0	A
UROP2100Q	Undergraduate Research Opportunities Series 2	1.0	1.0	P

TGA: 3.975
CGA: 3.832

Cumulative Credits Earned: 86.0
Cumulative Transfer Credits: 6.0

Awards: Dean's List
University's Scholarship Scheme for Continuing Undergraduate Students

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2025-26 Winter (Full-Time)

Awards: University's Scholarship Scheme for Continuing Undergraduate Students

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2025-26 Spring (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
COMP1942	Exploring and Visualizing Data	3.0	3.0	A
COMP1991	Industrial Experience	0.0	-	P
COMP4121	Modern Compiler Construction	3.0	3.0	A
COMP4551	Large-Scale Machine Learning System for Foundation Models	3.0	3.0	A-
COMP4900	Academic and Professional Development	0.0	-	PP
ELEC5140	Advanced Computer Architecture	3.0	3.0	A
PHYS1002	Introduction to Astrophysics and Astronomy	3.0	3.0	A+

TGA: 4.000
CGA: 3.859

Cumulative Credits Earned: 101.0
Cumulative Transfer Credits: 6.0

Awards: Dean's List
University's Scholarship Scheme for Continuing Undergraduate Students

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2025-26 Summer (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
COMP4981	Final Year Project	6.0	-	PP

TGA: **
CGA: 3.859

Cumulative Credits Earned: 101.0
Cumulative Transfer Credits: 6.0

Awards: University's Scholarship Scheme for Continuing Undergraduate Students

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- End of Transcript -

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EXPLANATORY NOTES

Academic Transcript and Release of Records
The academic transcript provides the full record of a student's academic results and performance. This record may not be released to any other party without the written consent of the student, in accordance with Personal Data (Privacy) Ordinance established in 1996.

Calendar
The academic year begins on 1 September and ends on 31 August of the following year. There are four terms in each academic year, namely Fall, Winter, Spring and Summer. Fall and Spring are regular terms with thirteen weeks of scheduled classes.

Degree Awarded
The undergraduate programs offered by the University all lead to honors degrees. The University also offers postgraduate programs leading to master degrees, doctoral degrees and graduate diplomas.

Credit Unit
Credit units are earned through the completion of courses and other assessed learning activities. For undergraduate courses, one credit unit generally represents 40 to 50 learning hours. For postgraduate courses, one credit is normally designated for one lecture, one tutorial or three laboratory/field study hours per week.

Grading
Students receive a grade in each course in which they are enrolled. Transfer credits and courses graded HP/P/LP/U/Y, AU, CR, DI, DN, I, IP, P, PA, PP, PS ,T and W are omitted from the calculation of grade point averages.

Letter Grade	Definition for Undergraduate Course	Definition for Postgraduate Course
A+, A, A-	Excellent Performance	Excellent Performance
B+, B	Good Performance	Good Performance
B-	Good Performance	Marginal Performance ①
C+, C	Satisfactory Performance	Marginal Performance
C-	Satisfactory Performance	–
D	Marginal Pass	–
F	Failure	Failure

Other Grade	Definition
AU	Audited
CR	Credit
DI	Distinction
DN	Distinction
I	Incomplete
IP	In Progress ②
P	Pass, Ungraded
PA	Pass, Graded
PP	Permitted to Proceed
PS	Pass, Graded
T	Transferred
W	Withdrawal without Penalty

HP/P/LP/U/Y Grade (used in Executive Programs)	Definition
HP	High Pass
P	Pass
LP	Low Pass
U	Unsatisfactory
Y	Incomplete

① Definition revised from 'Good Performance' from 1999-2000 Fall
② Ceased to be used from 1997-98 Fall

Grade Averages
With effect from 2010-11 Spring, the University adopted a 4-point grading system to replace the 12-point system that had been used since 1991-92 in the calculation of grade averages. The following grade averages are calculated:

TGA (Term Grade Average): TGA is calculated across all courses taken in a term.

CGA (Cumulative Grade Average): CGA is calculated for all the courses earning grade points completed since first registration in a program up to the point of calculation.

GGA (Graduation Grade Average): GGA is the student's CGA adjusted by excluding courses from the calculation under established University rules.

MCGA (Major Cumulative Grade Average) is calculated based on the grade points earned for the courses used to fulfill the major requirements and the requirements of the corresponding School or Academy of Interdisciplinary Studies, where appropriate. For students admitted in 2018-19 academic year or after, the classification of awards is based on a two-tier system of MCGA and overall CGA.

Before 2010-11 Spring, SGA (Semester Grade Average) was used:
SGA (Semester Grade Average) is the combined grade average covering all courses taken in Fall including the following Winter or Spring including the following Summer.

The following table shows the conversion of 12-point and 4-point grading system:

12-point System (1991-92 Fall to 2010-11 Winter) Course Grade Point	Letter Grade	4-point System (effective from 2010-11 Spring) Course Grade Point
12	A+	4.3
11	A	4.0
10	A-	3.7
9	B+	3.3
8	B	3.0
7	B-	2.7
6	C+	2.3
5	C	2.0
4	C-	1.7
-	D+	-
2	D	1.0
-	D-	-
0	F	0

Further information about the academic transcript is available at https://registry.hkust.edu.hk/transcript. Please email all queries to the Academic Registry of the University at artranscript@ust.hk.

The Hong Kong University of Science and Technology issues some official documents in electronic format which are verifiable. The Blockcerts technology is adopted in registering and verifying the electronic documents in the format of a JSON file.

The JSON file and the associated URL are issued to the graduates/students who may pass them to other parties at their discretion. **After successful verification of the JSON file at the University official website (https://registry.hkust.edu.hk/verify), an authenticated copy of the document can be viewed or downloaded in formats such as PDF or HTML.**